Project in Preparation of the Master's Thesis

The Quark DSE and Pion BSE in the Maris-Tandy model

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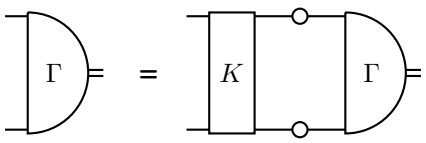
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Chapter 1

Introduction

Chapter 2

Theory





(2.1)

2.1 Quark DSE

2.2 Pion BSE

$$G = G_0 + G_0 T G_0 \tag{2.2}$$

$$T = K + K G_0 T \tag{2.3}$$

$$T \propto \frac{\Gamma\bar{\Gamma}}{P^2 + m^2} \tag{2.4}$$

$$\Gamma = K G_0 \Gamma \tag{2.5}$$

$$[\Gamma(p,P)]_{tu} = \int \mathrm{d}^4 q \, [S(q_+)\Gamma(q,P)S(q_-)]_{sr} \, K^{rs}_{tu}(q,k,P) \tag{2.6}$$

$q_\pm = q + \eta_\pm P$ $q_\pm = q \pm \frac{1}{2}P$

2.3 Herleitung Esther

$$G^{(n)}[\phi] = \langle\Omega|\,\mathrm{T}\,\{\phi(x_1)\phi(x_2)...\phi(x_n)\}\,|\Omega\rangle \tag{2.7}$$

$$\langle\Omega|\,\mathrm{T}\,\phi(x)\phi(y)\,|\Omega\rangle \equiv \begin{cases} \langle\Omega|\,\phi(x)\phi(y)\,|\Omega\rangle\,, & \text{for } x^0 > y^0 \\ \langle\Omega|\,\phi(y)\phi(x)\,|\Omega\rangle\,, & \text{for } y^0 > x^0 \end{cases} \tag{2.8}$$

$$G^{(n)}[\phi] = \frac{\int \mathcal{D}\phi\ \phi(x_1)...\phi(x_n)e^{-S[\phi]}}{\int \mathcal{D}\phi\ e^{-S[\phi]}} \tag{2.9}$$

$$Z[J] = \int \mathcal{D}\phi\ e^{-S[\phi]+J\phi} \tag{2.10}$$

$$G[\phi] = \langle\Omega|\,\mathrm{T}\,G[\phi]\,|\Omega\rangle = \left[G\left[\frac{\delta}{\delta J}\right]\frac{Z[J]}{Z[0]}\right]_{J=0} \tag{2.11}$$

$$Z[J] = e^{-W[J]} = e^{-(\Gamma[\tilde{\phi}]-\int_x \tilde{\phi}J(x))}, \quad \Gamma[\tilde{\phi}] = W[J] - \tilde{\phi}J \tag{2.12}$$

$$\tilde{\phi}(x) = -\frac{\delta W[J]}{\delta J(x)} = \frac{1}{Z[J]}\frac{\delta}{\delta J(x)}Z[J] = \langle\phi(x)\rangle \tag{2.13}$$

$$S_{\mathrm{QCD}} = \int \mathrm{d}^4x\ \left(\bar{\psi}(\not{D}+m)\psi + \tfrac{1}{4}F^a_{\mu\nu}F_a^{\mu\nu}\right) \tag{2.14}$$

$$\psi' = U\psi, \quad A'_\mu = UA_\mu U^\dagger - \frac{\mathrm{i}}{g}U\partial_\mu U^\dagger, \quad F'_{\mu\nu} = UF_{\mu\nu}U^\dagger \tag{2.15}$$

$$\frac{\delta S[\phi]}{\delta\phi} = 0 \tag{2.16}$$

$$Z[J] = \int \mathcal{D}\phi\ e^{-S[\phi]+J\phi} \tag{2.17}$$

$$\int \mathcal{D}\phi\ e^{-S[\phi]+J\phi} = \int \mathcal{D}\phi\ e^{-S[\phi+\delta\phi]+J(\phi+\delta\phi)} \tag{2.18}$$

$$\left\langle\frac{\delta S[\phi]}{\delta\phi(x)}\right\rangle_J = J(x) \tag{2.19}$$

$$\langle f(\phi)\rangle_J \equiv Z[J]^{-1}\int \mathcal{D}\phi\; f(\phi)\, e^{-S[\phi]+J\phi} = Z[J]^{-1} f\left(\frac{\delta}{\delta J}\right) Z[J] \tag{2.20}$$

$$\left\langle \frac{\delta S[\phi]}{\delta\phi(x)} \right\rangle = Z[J]^{-1}\frac{\delta S}{\delta\phi}\left[\frac{\delta}{\delta J(x)}\right] Z[J] = J(x) \tag{2.21}$$

$$\frac{\delta\Gamma}{\delta\tilde{\phi}(x)} = \frac{\delta S}{\delta\phi}\left[\tilde{\phi}(x) + \int_y \frac{\delta^2\Gamma}{\delta\tilde{\phi}(x)\delta\tilde{\phi}(y)}\frac{\delta}{\delta\tilde{\phi}(y)}\right] \tag{2.22}$$

$$S^{-1} = S_0^{-1} + \int g\gamma_\mu D^{\mu\nu} S\,\Gamma_\nu \tag{2.23}$$

$$S(p) = G^{(2)}(p) = \int \mathrm{d}^4x\; e^{-\mathrm{i}px}\,\langle\Omega|\,\mathrm{T}\,\bar{\psi}(x)\,\psi(0)\,|\Omega\rangle \tag{2.24}$$

$$S_{\alpha\beta abAB}(p) = S_{\alpha\beta}(p)\otimes\delta^{ab}\otimes\delta^{AB} \tag{2.25}$$

$$S_{\alpha\beta}(p) = -\mathrm{i}\sigma_v(p)\not{p} + \sigma_s(p)\mathbb{1}_D \tag{2.26}$$

$$S^{-1}(p) = \mathrm{i}\not{p}A(p^2) + B(p^2) \tag{2.27}$$

$$S^{-1}(p^2) = S_0^{-1}(p^2) + \int \mathrm{d}^4q\; D_{ab}^{\mu\nu}(p-q)\left(g\frac{\lambda^a}{2}\gamma_\mu\right) S(q^2)\left(g\Gamma_\nu^b(p,q)\right) \tag{2.28}$$

$$m_0(\Lambda^2) = Z_m(\mu^2,\Lambda^2)\, m(\mu^2) \tag{2.29}$$

$$S(p^2,\Lambda^2) = Z_2(\mu^2,\Lambda^2)\, S_R(p^2,\mu^2) \tag{2.30}$$

$$A^{-1}(p^2,\Lambda^2) = Z_2(\mu^2,\Lambda^2)\, A_R^{-1}(p^2,\mu^2) \tag{2.31}$$

$$Z_{1F} = \frac{Z_2}{\tilde{Z}_3}, \quad Z_g\tilde{Z}_3 Z_3^{1/2} = 1 \tag{2.32}$$

$$S^{-1}(p^2,\mu^2) = Z_2(\mathrm{i}\not{p} + Z_m m) \tag{2.33}$$

$$+ Z_g^2 g^2 \int \mathrm{d}^4 q \; Z_3 D_{ab}^{\mu\nu}(p-q,\mu^2) \left(\frac{\lambda^a}{2}\gamma_\mu\right) Z_2^2 S(q^2,\mu^2) \left(Z_{1F}^{-1}\Gamma_\nu^b(p,q,\mu^2)\right) \tag{2.34}$$

$$g(\mu^2) = Z_g^{-1}(\mu^2,\Lambda^2) g(\Lambda^2) \tag{2.35}$$

$$Z_g^2 Z_3 Z_2^2 \frac{\tilde{Z}_3}{Z_2} = \frac{Z_2}{\tilde{Z}_3} \cdot \left[Z_g^2 \tilde{Z}_3^2 Z_3\right] = \frac{Z_2}{\tilde{Z}_3} = Z_{1F} \tag{2.36}$$

$$S^{-1}(p^2,\mu^2) = Z_2(\mathrm{i}\not{p} + Z_m m) \tag{2.37}$$

$$+ \frac{Z_2}{\tilde{Z}_3} g^2 \int \mathrm{d}^4 q \; D_{ab}^{\mu\nu}(p-q,\mu^2) \left(\frac{\lambda^a}{2}\gamma_\mu\right) S(q^2,\mu^2) \left(\Gamma_\nu^b(p,q,\mu^2)\right) \tag{2.38}$$

$$A(\mu^2) \stackrel{!}{=} 1, \quad B(\mu^2) \stackrel{!}{=} m(\mu^2) \tag{2.39}$$

$$Z_2 = 1 - \Sigma_A(\mu^2,\Lambda^2), \quad Z_4 = 1 - \frac{\Sigma_B(\mu^2,\Lambda^2)}{m(\mu^2)} \tag{2.40}$$

$$\frac{Z_2}{\tilde{Z}_3} g^2 D_{ab}^{\mu\nu}(p-q)\Gamma_\nu^b(p,q) \to Z_2^2 \mathcal{G}\left((p-q)^2\right) \frac{t^{\mu\nu}(p-q)}{(p-q)^2} \frac{\lambda^a}{2}\gamma_\nu \tag{2.41}$$

$t^{\mu\nu}(k) = \delta^{\mu\nu} - \frac{k^\mu k^\nu}{k^2}$

$$\frac{\mathcal{G}(k^2)}{4\pi} = \alpha(k^2) = \frac{D}{\omega^6}\pi k^4 e^{-\frac{k^2}{\omega^2}} + \frac{2\pi\gamma_m\left(1-e^{-k^2}\right)}{\ln\left[e^2 - 1 + \left(1+\frac{k^2}{\Lambda_{\mathrm{QCD}}^2}\right)^2\right]} \tag{2.42}$$

$$\frac{\alpha(k^2)}{k^2} \to \frac{\alpha(k^2)}{k^2}\left(\frac{\Lambda_{\mathrm{PV}}^2}{k^2+\Lambda_{\mathrm{PV}}^2}\right) \tag{2.43}$$

$$A(p^2) = Z_2 + \frac{16\pi}{3} Z_2^2 \int \mathrm{d}^4 q \; \frac{A(q^2)}{q^2A^2(q^2)+B^2(q^2)} \left[(pq) + 2\frac{(pk)(qk)}{k^2}\right] \frac{\alpha(k^2)}{k^2} \frac{\Lambda_{\mathrm{PV}}^2}{\Lambda_{\mathrm{PV}}^2+k^2} \tag{2.44}$$

$$B(p^2) = Z_4 m + 16\pi Z_2^2 \int \mathrm{d}^4 q \; \frac{B(q^2)}{q^2A^2(q^2)+B^2(q^2)} \frac{\alpha(k^2)}{k^2} \frac{\Lambda_{\mathrm{PV}}^2}{\Lambda_{\mathrm{PV}}^2+k^2} \tag{2.45}$$

$$\alpha(k^2) = \alpha_{\mathrm{UV}}(k^2) + \alpha_{\mathrm{IR}}(k^2) \tag{2.46}$$

$$\alpha_{\mathrm{IR}}(k^2) = \frac{D}{\omega^6}\pi k^4 e^{-\frac{k^2}{\omega^2}} \tag{2.47}$$

$$\alpha_{\mathrm{UV}}(k^2) = \frac{2\pi\gamma_m\left(1-e^{-k^2}\right)}{\ln\left[e^2 - 1 + \left(1+\frac{k^2}{\Lambda_{\mathrm{QCD}}^2}\right)^2\right]} \tag{2.48}$$

2.4 Herleitung Stephan

$$S_{\mathrm{QCD}} = \int \mathrm{d}^4x \left[\sum_{i=1}^{N_f} \overline{\psi}_i (\not{D} + m_i)\psi_i + \tfrac{1}{4} F^a_{\mu\nu} F_a^{\mu\nu} \right] \tag{2.49}$$

$$S^{-1}(p) = Z_2 S_0^{-1}(p) - \Sigma(p) \tag{2.50}$$

$$\Sigma(p) = -Z_{1F}(\mu^2, \Lambda^2) g^2 C_f \int \mathrm{d}^4q \; \gamma_\mu S(q) \Gamma_\nu(p,q) D^{\mu\nu}(p-q) \tag{2.51}$$

$$D^{\mu\nu}(k) = T^{\mu\nu}(k) \frac{Z(k^2)}{k^2}, \quad T^{\mu\nu}(k) = \delta^{\mu\nu} - k^\mu k^\nu / k^2 \tag{2.52}$$

$$\Gamma^\mu_{\mathrm{RL}}(p,q) = Z_{1F} \gamma^\mu \lambda_1(k^2) \tag{2.53}$$

$$\Sigma(p) = -C_f Z_{1F}^2 g^2 \int \mathrm{d}^4q \; \gamma_\mu S(q) \gamma_\nu T^{\mu\nu} \frac{\lambda_1(k^2) Z(k^2)}{k^2} \tag{2.54}$$

$$\alpha(k^2) := \left(\frac{\tilde{Z}_1}{\tilde{Z}_3} \right)^2 \frac{g^2}{4\pi} Z(k^2) \lambda_1(k^2) \tag{2.55}$$

$$Z_{1F}^2 = Z_2^2 \left(\frac{\tilde{Z}_1}{\tilde{Z}_3} \right)^2 \tag{2.56}$$

$$\Sigma(p) = -4\pi C_f Z_2^2 \int \mathrm{d}^4q \; \gamma_\mu S(q) \gamma_\nu T^{\mu\nu} \frac{\alpha(k^2)}{k^2} \tag{2.57}$$

$$S^{-1}(p) = Z_2 \left(\mathrm{i}\not{p} + Z_m m_c \right) + 4\pi C_f Z_2^2 \int \mathrm{d}^4q \; \gamma_\mu S(q) \gamma_\nu T^{\mu\nu} \frac{\alpha(k^2)}{k^2} \tag{2.58}$$

$C_f = \frac{4}{3}$

$$S^{-1}(p) = \mathrm{i}\not{p} A(p^2) + B(p^2) \tag{2.59}$$

$$A(p^2) = Z_2 + \Sigma_A(p^2) \tag{2.60}$$

$$B(p^2) = Z_2 Z_m m_c + \Sigma_B(p^2) \tag{2.61}$$

$$\Sigma_A(p^2) = \frac{16\pi}{3} Z_2^2 \int \mathrm{d}^4 q \, \frac{A(q^2)}{q^2 A^2(q^2) + B^2(q^2)} \left[(pq) + 2\frac{(pk)(qk)}{k^2} \right] \frac{\alpha(k^2)}{k^2} \frac{\Lambda_{\mathrm{PV}}^2}{\Lambda_{\mathrm{PV}}^2 + k^2} \tag{2.62}$$

$$\Sigma_B(p^2) = 16\pi Z_2^2 \int \mathrm{d}^4 q \, \frac{B(q^2)}{q^2 A^2(q^2) + B^2(q^2)} \frac{\alpha(k^2)}{k^2} \frac{\Lambda_{\mathrm{PV}}^2}{\Lambda_{\mathrm{PV}}^2 + k^2} \tag{2.63}$$

$$\alpha(k^2) = \alpha_{\mathrm{UV}}(k^2) + \alpha_{\mathrm{IR}}(k^2) \tag{2.64}$$

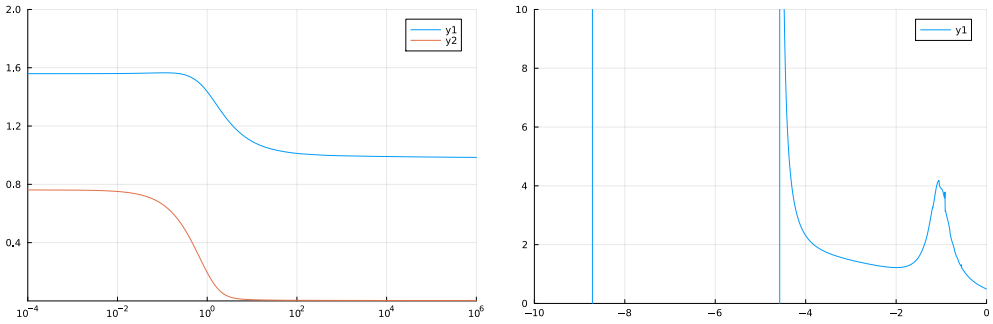
$$\alpha_{\mathrm{IR}}(k^2) = \frac{D}{\omega^6} \pi k^4 e^{-\frac{k^2}{\omega^2}} \tag{2.65}$$

$$\alpha_{\mathrm{UV}}(k^2) = \frac{2\pi\gamma_m \left(1 - e^{-k^2}\right)}{\ln\left[e^2 - 1 + \left(1 + k^2/\Lambda_{\mathrm{QCD}}^2\right)^2\right]} \tag{2.66}$$

Chapter 3

Results

3.1 Quark DSE



(a) real p (b) imaginary p

Figure 3.1: Quark amplitude dependence on real and imaginary momenta

3.2 Pion BSE

$$\Gamma_{\mathrm{PS}}(p,P) = \gamma^5 \left(E(p,P) - \mathrm{i}\not{P} F(p,P) - \mathrm{i}(p\cdot P)\not{p}\, G(p,P) + \left[\not{p},\not{P}\right]_- H(p,P) \right) \tag{3.1}$$

$$\Gamma(p,P) = -C_f Z_2^2 4\pi \int \mathrm{d}^4 q \gamma^\mu S(q_+)\Gamma(q,P)S(q_-)\gamma^\nu T_{\mu\nu}(k)\frac{\alpha(k^2)}{k^2} \tag{3.2}$$

Inserting our ansatz, applying projector p_E, and taking the Dirac trace, we get:

$$\begin{aligned} E(p,P) = \operatorname{tr}\Big[&- C_f Z_2^2 4\pi \frac{1}{4}\gamma^5 \int \mathrm{d}\!\!\!^{4} q \gamma^\mu S(q_+)\gamma^5 \big(E(q,P) - \mathrm{i}\not{P} F(q,P) \\ &- \mathrm{i}(q\cdot P)\not{q} G(q,P) + \big[\not{q},\not{P}\big]_- H(q,P)\big) S(q_-)\gamma^\nu T_{\mu\nu}(k)\frac{\alpha(k^2)}{k^2}\Big] \qquad (3.3) \\ = &- C_f Z_2^2 \pi \int \mathrm{d}\!\!\!^{4} q \operatorname{tr}\Big[\gamma^5\gamma^\mu S(q_+)\gamma^5 \big(E(q,P) - \mathrm{i}\not{P} F(q,P) \\ &- \mathrm{i}(q\cdot P)\not{q} G(q,P) + \big[\not{q},\not{P}\big]_- H(q,P)\big) S(q_-)\gamma^\nu T_{\mu\nu}(k)\Big]\frac{\alpha(k^2)}{k^2} \qquad (3.4) \end{aligned}$$

Observing the alone-standing trace:

$$\operatorname{tr}\left[\gamma^5\gamma^\mu \frac{\mathrm{i}\not{q}_+ A(q_+^2) + B(q_+^2)}{q_+^2 A^2(q_+^2) + B^2(q_+^2)}\gamma^5 \big(E(q,P) - \mathrm{i}\not{P} F(q,P) - \mathrm{i}(q\cdot P)\not{q} G(q,P) + \big[\not{q},\not{P}\big]_- H(q,P)\big) \frac{\mathrm{i}\not{q}_- A(q_-^2) + B(q_-^2)}{q_-^2 A^2(q_-^2) + B^2(q_-^2)}\gamma^\nu T_{\mu} \right. \qquad (3.5)$$

$$\begin{pmatrix} E(p) \\ F(p) \\ G(p) \\ H(p) \end{pmatrix} = \begin{pmatrix} \kappa_{EE}(p,q) & \kappa_{EF}(p,q) & \kappa_{EG}(p,q) & \kappa_{EH}(p,q) \\ \kappa_{FE}(p,q) & \kappa_{FF}(p,q) & \kappa_{FG}(p,q) & \kappa_{FH}(p,q) \\ \kappa_{GE}(p,q) & \kappa_{GF}(p,q) & \kappa_{GG}(p,q) & \kappa_{GH}(p,q) \\ \kappa_{HE}(p,q) & \kappa_{HF}(p,q) & \kappa_{HG}(p,q) & \kappa_{HH}(p,q) \end{pmatrix} \begin{pmatrix} E(q) \\ F(q) \\ G(q) \\ H(q) \end{pmatrix} \qquad (3.6)$$

$$\lambda(P^2) \begin{pmatrix} E \\ F \\ G \\ H \end{pmatrix} (p,P^2) = \mathcal{K}(p,q,P^2) \begin{pmatrix} E \\ F \\ G \\ H \end{pmatrix} (p,P^2) \qquad (3.7)$$

$$\lambda(-m_\pi^2) \stackrel{!}{=} 1 \qquad (3.8)$$

Projectors:

$$p_E = \frac{1}{4}\gamma^5 \qquad (3.9)$$

$$p_F = \frac{\mathrm{i}}{4\left((p\cdot P)^2 - p^2P^2\right)}\gamma^5 \left(p^2\not{P} - (p\cdot P)\not{p}\right) \qquad (3.10)$$

$$p_G = \frac{\mathrm{i}}{4\left((p\cdot P)^2 - p^2P^2\right)}\gamma^5 \left(\frac{P^2}{p\cdot P}\not{p} - \not{P}\right) \qquad (3.11)$$

$$p_H = \frac{1}{16\left((p\cdot P)^2 - p^2P^2\right)}\gamma^5 \left[\not{p},\not{P}\right]_- \qquad (3.12)$$

Chapter 4

Conclusion and Outlook

Appendix A

Mathematical Relations

A.1 Dirac algebra

The Dirac algebra is the algebra of the gamma matrices γ^μ used in the Dirac equation. It is defined by the anticommutation relation

$$\{\gamma^\mu, \gamma^\nu\} = 2\,\eta^{\mu\nu}\,\mathbf{1}, \tag{A.1}$$

where $\eta^{\mu\nu}$ is the defining metric of the space for which the gamma matrices are formulated, or the signature of the dot product of two four vectors. This ensures the Lorentz covariance of the equation. The gamma matrices are represented by four 4×4 matrices and generate the Clifford Algebra $\mathrm{Cl}_{1,3}(\mathbb{C})$. It is also customary to define a fifth matrix, γ^5, which satisfies the following relations:

$$\{\gamma^5, \gamma^\mu\} = 0\,, \qquad \gamma^5 = (\gamma^5)^\dagger\,, \qquad (\gamma^5)^2 = \mathbf{1} \tag{A.2}$$

Independent of the representation, we define the Feynman-Slash operator to be:

$$\not{p} = \gamma_\mu p^\mu$$

A.1.1 Minkowski representation

The metric of Minkowski space is given by the signature $(+,-,-,-)$, written as a matrix:

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \tag{A.3}$$

One representation of the gamma matrices that satisfies the Dirac algebra with this metric is

$$\gamma^0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \gamma^n = \begin{pmatrix} 0 & \sigma_n \\ -\sigma_n & 0 \end{pmatrix}, \tag{A.4}$$

where every 0 represents an empty 2×2 matrix and every 1 represents the 2×2 identity matrix. The σ_n serve as placeholders for the three Pauli matrices:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix} \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{A.5}$$

The fifth gamma matrix can now be calculated using the equation:

$$\gamma^5 = \gamma_5 = \mathrm{i}\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{A.6}$$

Furthermore we make use of the charge conjugation operator C, which possesses properties independent of the representation in which it is given, namely:

$$C^{-1}\gamma_\mu C = -\gamma_\mu^T\,, \qquad C\gamma_5 C^{-1} = \gamma_5^T \tag{A.7}$$

In Minkowski space, it is represented by the matrix

$$C = -C^T = -C^\dagger = \begin{pmatrix} 0 & \mathrm{i}\sigma_2 \\ \mathrm{i}\sigma_2 & 0 \end{pmatrix}. \tag{A.8}$$

A.1.2 Euclidean representation

When working in Euclidean space, it is customary to switch the indices 0 and 4. This also serves to differentiate clearly between four vectors in both representations. The space-like components keep the indices 1 through 3. Since the defining metric in Euclidean space is just the identity, the new gamma matrices need to satisfy

$$\{\gamma^\mu, \gamma^\nu\} = 2\,\delta^{\mu\nu}\,\mathbf{1}, \tag{A.9}$$

$$\{\gamma^5, \gamma^\mu\} = 0\,. \tag{A.10}$$

They can be computed with

$$\gamma_{n,E} = -\mathrm{i}\,\gamma_{n,M}, \tag{A.11}$$

$$\gamma_{4,E} = \gamma_{0,M}, \tag{A.12}$$

where $n \in \{1, 2, 3\}$. These new representations now satisfy the anticommutation relations for this Euclidean signature.
Regarding γ_5 and C, both are unchanged by this new metric:

$$\gamma_{5,E} = -\gamma_{1,E}\,\gamma_{2,E}\,\gamma_{3,E}\,\gamma_{4,E} = \gamma_{5,M} \tag{A.13}$$

$$C_E = C_M = \gamma_{2,E}\,\gamma_{4,E} \tag{A.14}$$

In this work the Euclidean metric is used primarily. Due to this, the index E will be dropped and it is assumed that every gamma matrix is used in its Euclidean representation unless stated otherwise.
One very important note is that when working with the Euclidean metric as defined above, the invariant square of a four vector is opposite in sign to its Minkowski counterpart. To counteract this change, all four vector products in Minkowski space need to change their sign when transferred to Euclidean space:

$$p_E^\mu q_{\mu,E} = -p_M^\mu q_{\mu,M} \tag{A.15}$$

A.2 Hyperspherical Coordinates

Over the course of this work it often becomes necessary to integrate over four dimensions. To facilitate this we utilize hyperspherical coordinates, with their general position vector being given by

$$p = |p| \begin{pmatrix} \cos\phi \sin\theta \sin\psi \\ \sin\phi \sin\theta \sin\psi \\ \cos\theta \sin\psi \\ \cos\psi \end{pmatrix}, \tag{A.16}$$

where $|p|$ represents the magnitude of the vector, and θ, ϕ, and ψ are the angles parameterizing its direction.
To integrate over this coordinate system, we calculate the Jacobian and get:

$$\int \mathrm{d}^4 p = \int_0^\infty \mathrm{d}p\, p^3 \int_0^{2\pi} \mathrm{d}\phi \int_0^\pi \mathrm{d}\theta\, \sin\theta \int_0^\pi \mathrm{d}\psi\, \sin^2\psi \tag{A.17}$$

Again we keep in mind the translation from Minkowski to Euclidean space, which necessitates us to bring in another factor i.

$$\int \mathrm{d}^4 p_M \to \mathrm{i} \int \mathrm{d}^4 p_E \tag{A.18}$$

Appendix B

Numerical Methods

B.1 Integration

B.2 Software